Trifluridine and Tipiracil (Lonsurf®) Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Treatment of adult patients with metastatic colorectal cancer (CRC) who have been previously treated with, or are not considered candidates for, available therapies including fluoropyrimidine-, oxaliplatin- and irinotecan-based chemotherapies, anti-VEGF agents, and anti-EGFR agents	C17-20	00382a	CDS: 01/02/2017

*This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patient's individual clinical circumstances.

The recommended starting dose of Lonsurf® in adults is 35 mg/m^2/dose administered orally twice daily on Days 1 to 5 and Days 8 to 12 of each cycle.

Cycle frequency and duration:

Each cycle is 28 days.
Treatment should continue until disease progression or unacceptable toxicity.

Day	Drug	Dose	Route	Cycle
1-5 **and** 8-12	Lonsurf®	35mg/m^2 BD* expressed in terms of trifluridine	PO within 1 hour after completion of morning and evening meals	All cycles
*The dosage must not exceed 80mg/dose See **Table 1**: Starting dose calculation according to body surface area.				
If doses are missed or held, the patient must not make up for missed doses				
Lonsurf ® is available as tablets containing either 15mg/6.14mg and 20mg/8.19mg of trifluridine and tipiracil (as hydrochloride) respectively				

Table 1: Starting dose calculation according to body surface area (BSA)

Starting Dose	BSA (m^2)	Dose in mg (x 2 daily)	Tablets per dose (X2 daily)		Total daily dose (mg)
			15mg/6.14mg	20mg/8.19mg	
35mg/m^2	< 1.07	35	1	1	70
	1.07-1.22	40	0	2	80
	1.23-1.37	45	3	0	90
	1.38-1.52	50	2	1	100
	1.53-1.68	55	1	2	110
	1.69-1.83	60	0	3	120
	1.84-1.98	65	3	1	130
	1.99-2.14	70	2	2	140
	2.15-2.29	75	1	3	150
	≥ 2.30	80	0	4	160

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ELIGIBILITY:

- Indication as above
- ECOG status 0-1
- Adequate haematological, hepatic and renal function

EXCLUSIONS:

- Hypersensitivity to trifluridine, tipiracil or any of the excipients
- Pregnancy*
  *See Supportive Care for further details
- Breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Proteinuria by dipstick analysis

Regular tests:

- FBC, renal and liver profile prior to each cycle or as clinically indicated
- Proteinuria by dipstick analysis prior to each cycle or as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant
- Dosing adjustments may be required based on individual safety and tolerability
- A maximum of 3 dose reductions are permitted to a minimum dose of 20mg/m^2 twice daily
- Dose escalation is not permitted after it has been reduced
- In the event of haematological and/or non-haematological toxicities patients should follow the dose interruption, resumption and reduction criteria stated in Table 2, Table 3 and Table 7

Haematological:

Table 2: Dose interruption and resumption criteria for haematological toxicities related to myelosuppression

Parameter	Interruption Criteria	Resumption Criteria[a]
ANC	< 0.5 × 10^9/L	≥ 1.5 × 10^9/L
Platelets	< 50 × 10^9/L	≥ 75 × 10^9/L

[a] Resumption criteria applied to the start of the next cycle for all patients regardless of whether or not the interruption criteria were met.

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Table 3: Recommended dose modifications for Lonsurf® in case of haematological adverse reactions

Adverse Reaction	Recommended Dose Modification
Febrile neutropenia	• Interrupt dosing until toxicity resolves to Grade 1 or baseline. • When resuming dosing, decrease the dose level by 5mg/m^2/dose from the previous dose level (Table 4) • Dose reductions are permitted to a minimum dose of 20mg/m^2/dose twice daily • Do not increase dose after it has been reduced.
ANC < 0.5 × 10^9/L or platelets < 25 × 10^9/L that results in more than 1 week's delay in start of next cycle	

Table 4: Dose reductions according to body surface area (BSA)

Reduced dose	BSA (m^2)	Dose in mg (x 2 daily)	Tablets per dose		Total daily dose (mg)
			15mg/6.14mg	20mg/8.19mg	
Level 1 dose reduction: From 35mg/m^2 to 30mg/m^2					
30mg/m^2	< 1.09	30	2	0	60
	1.09-1.24	35	1	1	70
	1.25-1.39	40	0	2	80
	1.40-1.54	45	3	0	90
	1.55-1.69	50	2	1	100
	1.70-1.94	55	1	2	110
	1.95-2.09	60	0	3	120
	2.10-2.28	65	3	1	130
	≥ 2.29	70	2	2	140
Level 2 dose reduction: From 30mg/m^2 to 25mg/m^2					
25mg/m^2	< 1.10	25 [a]	2[a]	1 [a]	50 [a]
	1.10-1.29	30	2	0	60
	1.30-1.49	35	1	1	70
	1.50-1.69	40	0	2	80
	1.70-1.89	45	3	0	90
	1.90-2.09	50	2	1	100
	2.10-2.29	55	1	2	110
	≥ 2.3	60	0	3	120
Level 3 dose reduction: From 25mg/m^2 to 20mg/m^2					
20mg/m^2	< 1.14	20	0	1	40
	1.14-1.34	25 [a]	2[a]	1 [a]	50 [a]
	1.35-1.59	30	2	0	60
	1.60-1.94	35	1	1	70
	1.95-2.09	40	0	2	80
	2.10-2.34	45	3	0	90
	≥ 2.35	50	2	1	100

[a] At a total daily dose of 50mg, patients should take 1 x 20mg/8.19mg tablet in the morning and 2 x 15mg/6.14mg tablets in the evening

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Renal and Hepatic Impairment:

Table 5: Dose modification of Lonsurf® in renal and hepatic impairment

Renal Impairment		Hepatic Impairment	
CrCl (mL/minute)	**Recommended dose**	**Severity**	**Recommended dose**
≥ 30	No dose adjustment is needed	**Mild**	No dose adjustment is needed
< 30	Reduce dose according to drug label, based on BSA (Refer to **Table 6** below)	**Moderate/ Severe**	Not recommended
Haemodialysis	Not recommended		
Renal and hepatic impairment dose modification recommendations obtained from Giraud et al. 2023			

Table 6: Starting dose and dose reduction in patients with severe renal impairment according to body surface area (BSA)

Reduced dose	BSA (m^2)	Dose in mg (x 2 daily)	Tablets per dose		Total daily dose (mg)
			15mg/6.14mg	20mg/8.19mg	
Starting dose					
20mg/m^2	< 1.14	20	0	1	40
	1.14–1.34	25[a]	2[a]	1[a]	50[a]
	1.35-1.59	30	2	0	60
	1.60-1.94	35	1	1	70
	1.95-2.09	40	0	2	80
	2.10-2.34	45	3	0	90
	≥ 2.35	50	2	1	100
Dose reduction: From 20mg/m^2 to 15mg/m^2					
15mg/m^2	< 1.15	15	1	0	30
	1.15-1.49	20	0	1	40
	1.50-1.84	25[a]	2[a]	1[a]	50[a]
	1.85-2.09	30	2	0	60
	2.10-2.34	35	1	1	70
	≥ 2.35	40	0	2	80

[a] At a total daily dose of 50mg, patients should take 1 x 20mg/8.19mg tablet in the morning and 2 x 15mg/6.14mg tablets in the evening

Management of adverse events:

Table 7: Dose Modification for Lonsurf® in case of adverse reaction

Adverse reactions	Recommended dose modification
Non-haematologic Grade 3 or Grade 4 adverse reaction; except for Grade 3 nausea and/or vomiting controlled by antiemetic therapy or diarrhoea responsive to anti-diarrhoeal medicinal products	• Interrupt dosing until toxicity resolves to Grade 1 or baseline. • When resuming dosing, decrease the dose level by 5 mg/m^2/dose from the previous dose level (Table 4) • Dose reductions are permitted to a minimum dose of 20mg/m^2/dose twice daily (or 15 mg/m^2/dose twice daily in severe renal impairment) • Do not increase dose after it has been reduced

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SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Trifluridine/tipiracil: Moderate to High **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None recommended

OTHER SUPPORTIVE CARE:

- Patients should be advised that Lonsurf® has minor influence on the ability to drive and use machines. Fatigue, dizziness or malaise may occur during treatment.
- Adequate contraception:
  - Women: Trifluridine may cause foetal harm when administered to pregnant women. Women should avoid becoming pregnant while taking Lonsurf® and for up to 6 months after ending treatment. Therefore, women of child-bearing potential must use highly effective contraceptive measures while taking Lonsurf® and for 6 months after stopping treatment. It is currently unknown whether Lonsurf® may reduce the effectiveness of hormonal contraceptives, and therefore women using hormonal contraceptives should add a barrier contraceptive method
  - Men with a partner of child-bearing potential must use effective contraception during treatment and for up to 6 months after discontinuation of treatment

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for full details of adverse effects.
- Please note the NCCP National SACT Regimens do not provide information regarding black triangle (▼) status. Please refer to the relevant Summary of Product Characteristics (SmPC) for this information.

REGIMEN SPECIFIC COMPLICATIONS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.
- **Gastrointestinal toxicity:** Lonsurf® caused an increase in the incidence of gastrointestinal toxicities including nausea, vomiting and diarrhoea. Patients with nausea, vomiting, diarrhoea and other gastrointestinal toxicities should be carefully monitored, and anti-emetic, anti-diarrhoeal and other measures, such as fluid/electrolyte replacement therapy, should be administered as clinically indicated. Dose modifications (delay and/or reduction) should be applied as necessary.
- **Lactose intolerance:** Lonsurf® contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

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- **Renal impairment:** patients with renal impairment should be monitored closely when being treated with Lonsurf®; patients with moderate or severe renal impairment should be more frequently monitored for haematological toxicities.

DRUG INTERACTIONS:

- Current drug interaction databases and SmPC should be consulted for more information.

REFERENCES:

1. Mayer R Van Cutsem E et al. Randomized Trial of TAS-102 for Refractory Metastatic Colorectal Cancer. N Engl J Med 2015;372:1909-19.
2. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
3. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
4. Trifluridine and tipiracil (Lonsurf®) film-coated tables. Summary of Product Characteristics (EMA). Last updated 11/03/2023. Accessed 25/11/2025. Available at: https://www.ema.europa.eu/en/medicines/human/EPAR/lonsurf#assessment-history

From 01/07/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

Version	Date	Amendment	Approved By
1	1/02/2017		Dr Maccon Keane
2	16/01/2019	Updated to new NCCP template. Updated details as per smpc update regarding drug strengths and rounding Updated dose modification in hepatic impairment as per SmPC. Updated drug interactions	Prof Maccon Keane
3	06/01/2021	Updated dose modification in renal impairment as per SmPC, amended drug interactions.	Prof. Maccon Keane
4	01/07/2026	Updated template, general formatting and grammar updates. NCCP Standardisation updates. Updated ICD-10 codes, renal and hepatic impairment to Giraud et al; updated emetogenic potential, updated regimen specific complications section, updated references. Updated regimen review schedule to 'as required'.	Prof. Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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